

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 13:26:02

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

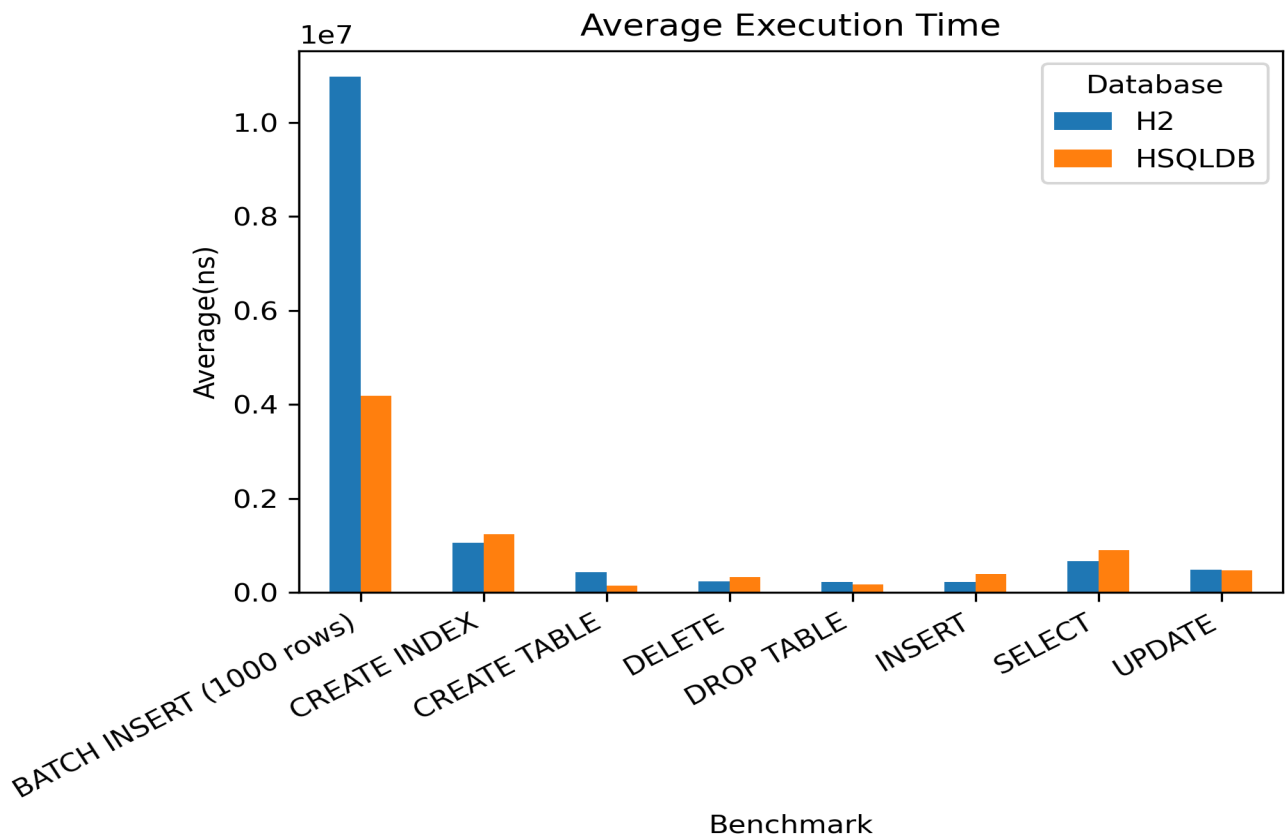
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

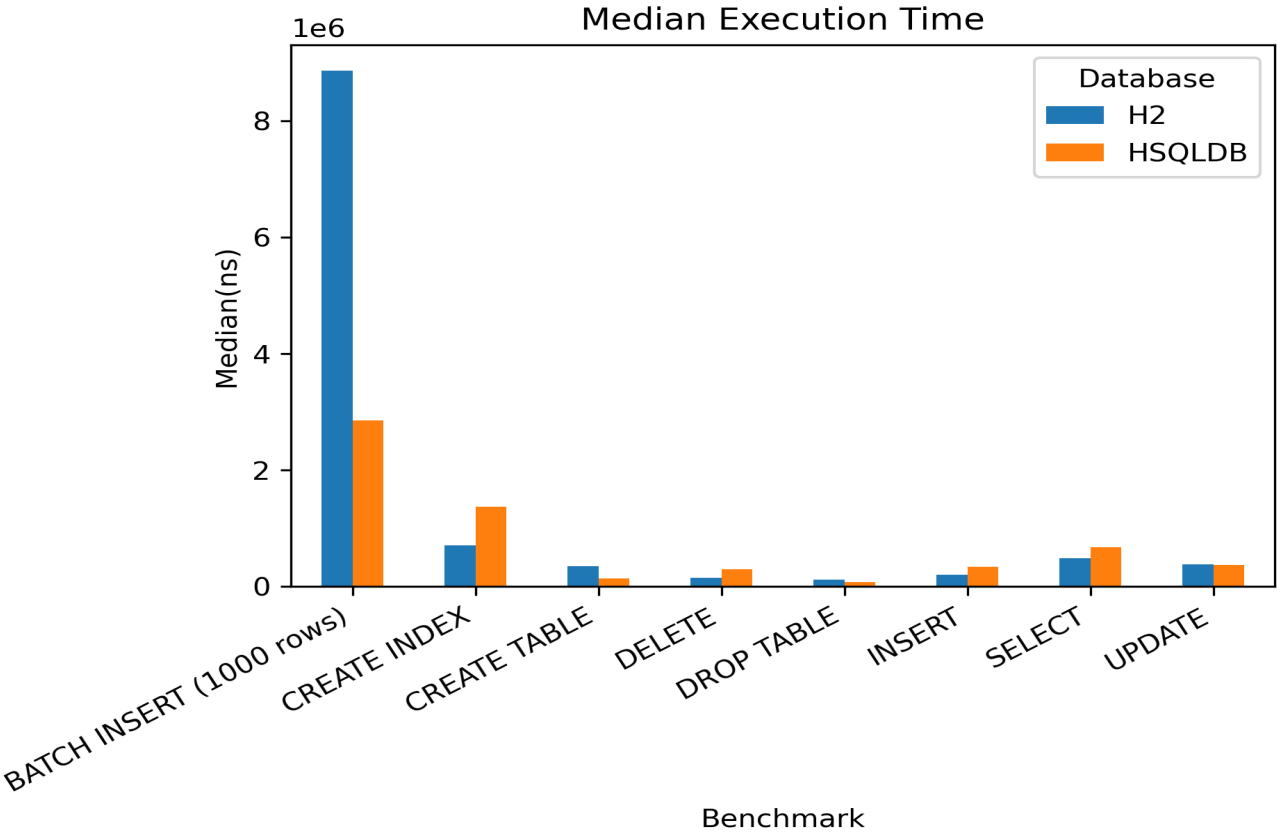
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	434494	353800	339863	261675	1886428
H2	INSERT	224779	200930	62495	169591	413003
H2	BATCH INSERT (1000 rows)	10974495	8867451	5645690	1874266	22787337
H2	SELECT	668811	483831	481984	318492	2028629
H2	UPDATE	477614	381868	417012	299747	2263901
H2	DELETE	233836	147118	283053	96703	1372005
H2	DROP TABLE	224522	121710	336971	109918	1641324
H2	CREATE INDEX	1058388	710915	726608	499535	3576324
HSQldb	CREATE TABLE	144783	134249	35380	116941	258108
HSQldb	INSERT	393350	334097	251108	248841	1432428
HSQldb	BATCH INSERT (1000 rows)	4183594	2858572	4706902	2356397	24512620
HSQldb	SELECT	905497	671655	375133	563275	1669858
HSQldb	UPDATE	469547	373316	395341	261174	2056549
HSQldb	DELETE	321940	300729	75874	231608	527067
HSQldb	DROP TABLE	166923	73479	362264	65083	1742716
HSQldb	CREATE INDEX	1235163	1367411	339727	326748	1489186

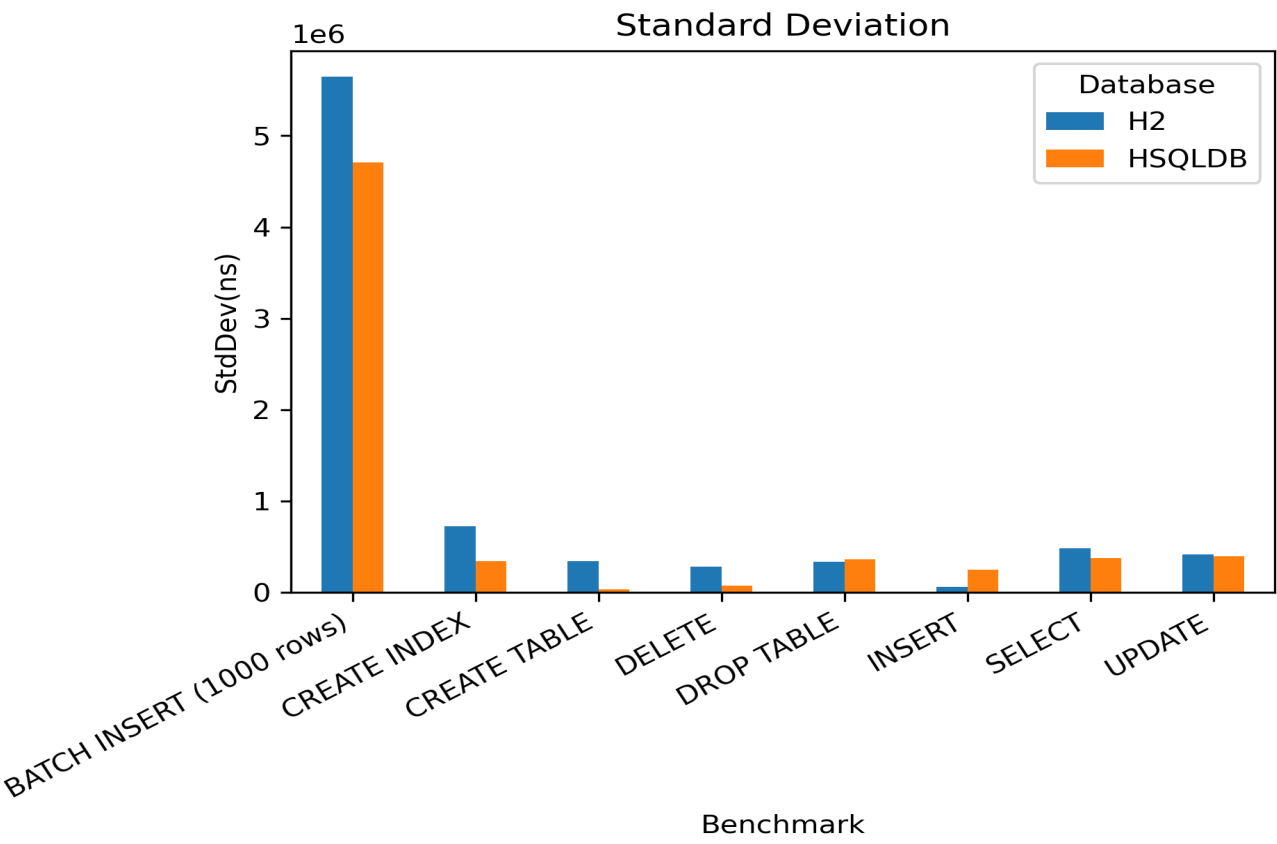
Average Execution Time



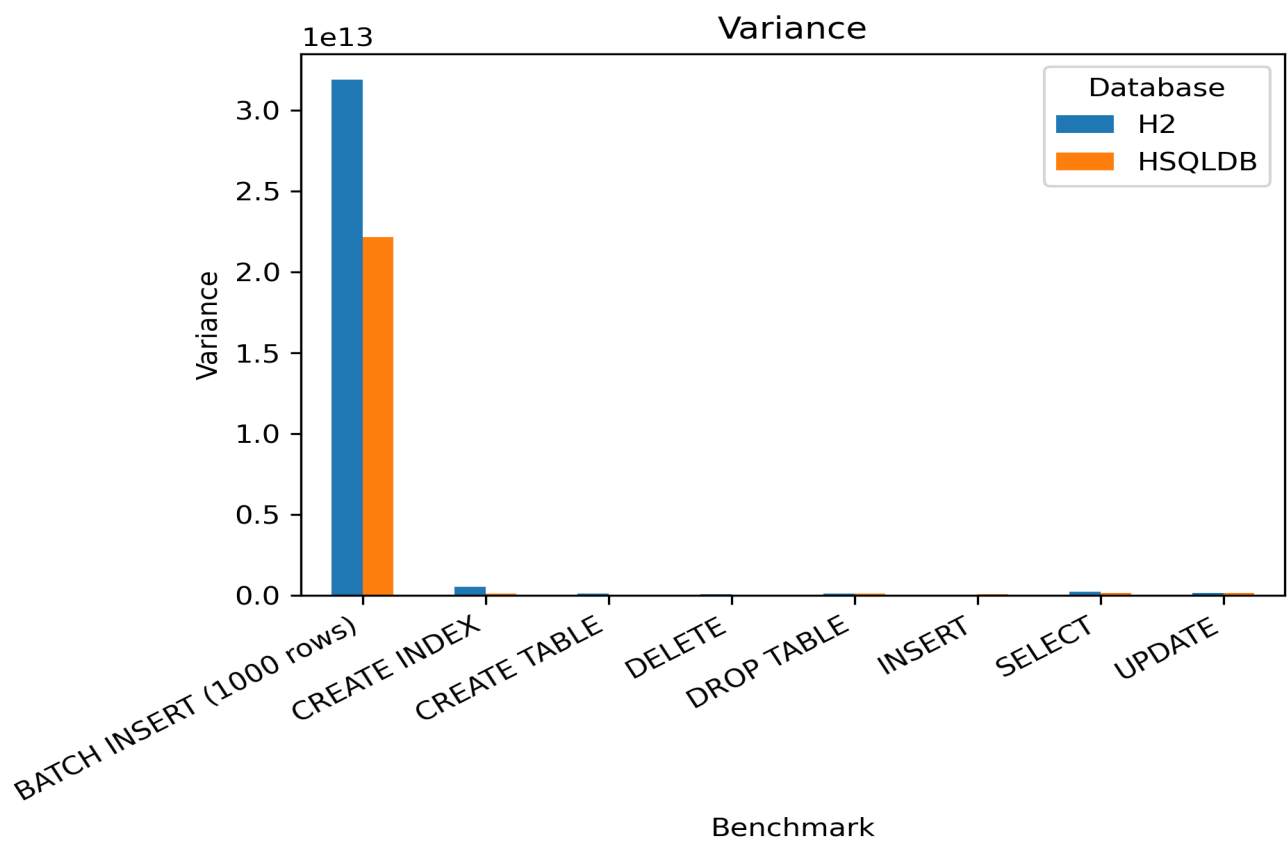
Median Execution Time



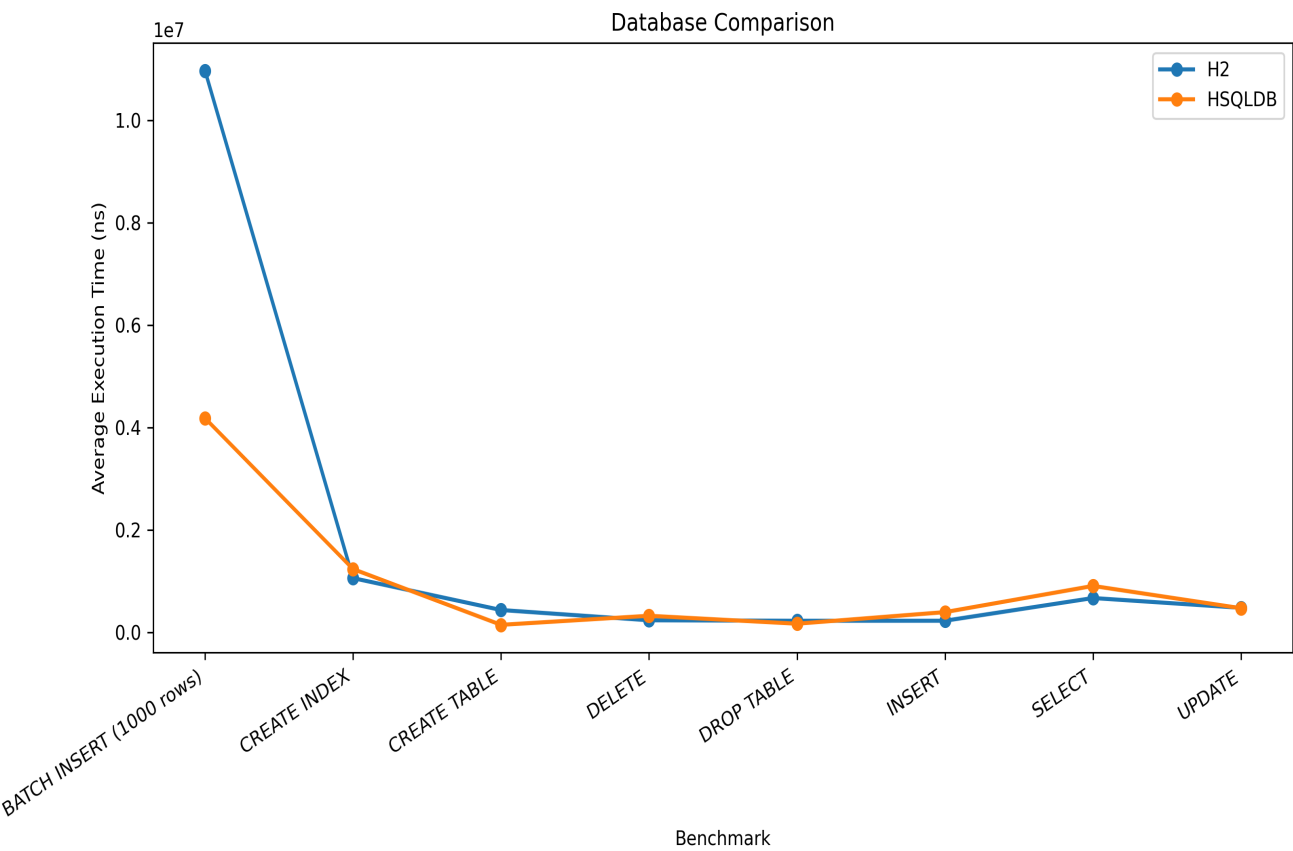
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	HSQLDB	4,183,594
CREATE INDEX	H2	1,058,388
CREATE TABLE	HSQLDB	144,783
DELETE	H2	233,836
DROP TABLE	HSQLDB	166,923
INSERT	H2	224,779
SELECT	H2	668,811
UPDATE	HSQLDB	469,547

Overall Benchmark Score

Database	Benchmarks Won
HSQLDB	4
H2	4

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	4.027	0.000271	YES
CREATE INDEX	-0.961	0.345277	NO
CREATE TABLE	3.696	0.001490	YES
DELETE	-1.311	0.203718	NO
DROP TABLE	0.507	0.614779	NO
INSERT	-2.840	0.009717	YES
SELECT	-1.689	0.099869	NO
UPDATE	0.061	0.951526	NO

Interpretation

A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **HSQldb** achieved the best overall performance, winning 4 benchmark categories.

The fastest benchmark measured was **CREATE TABLE** executed on **HSQldb** with an average execution time of **144,783 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **10,974,495 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.